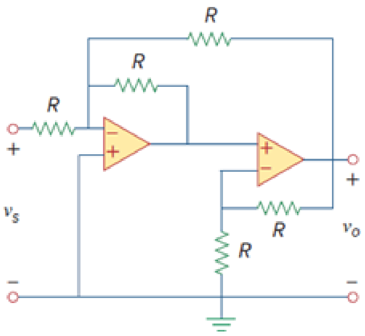
Chapter 5, Problem 54P

Problem

Determine the voltage transfer ratio v_o/v_s in the op amp circuit of Fig. 5.82, where R = 10 kΩ.

Figure. 5.82:



Step-by-step solution

Step 1 of 4

Consider the following circuit diagram:

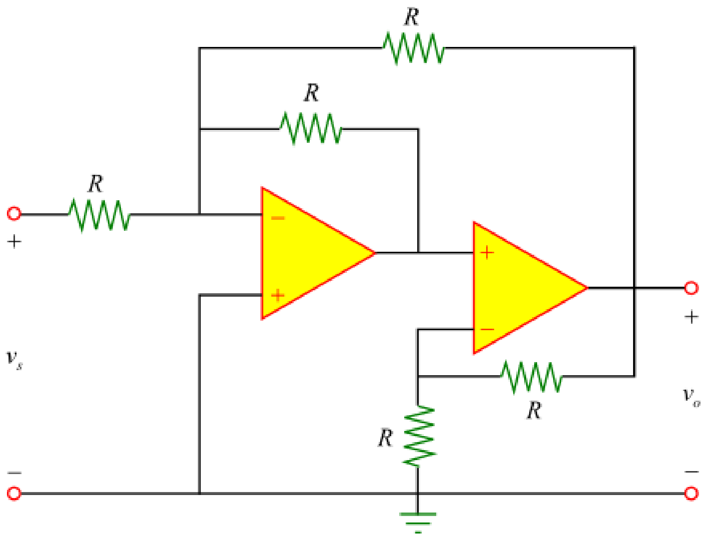


Figure 1

Step 2 of 4

Consider that the resistance, R in the circuit of Figure 1 is $10\ \text{k}\Omega$.

Assume the output of first stage as v_1.

Redraw the circuit in Figure 1 as follows:

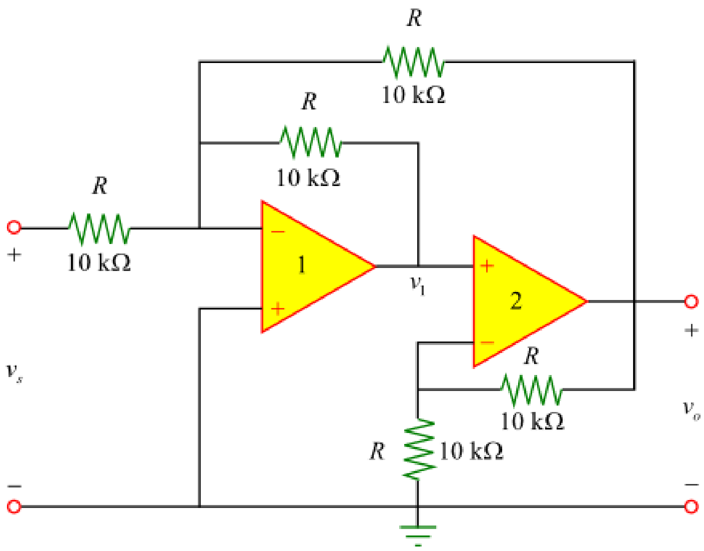


Figure 2

Step 3 of 4

The circuit in Figure 2 consists of a summing amplifier and a noninverting amplifier cascaded.

The summing amplifier combines several inputs and produces an output as a weighted sum of the inputs.

The summing amplifier in the circuit has two inputs v_s and v_o.

The following is the output of the summing amplifier (first operational amplifier).

$$v_1 = -\left(\frac{R}{R}v_s + \frac{R}{R}v_o\right)$$

Simplify this expression to get the following.

$$v_1 = -\left(v_s + v_o\right) \quad \cdots\cdots (1)$$

Step 4 of 4

The noninverting amplifier provides a positive voltage gain of $\left(1+\frac{R_f}{R_1}\right)$.

The following is the output of the noninverting amplifier (second operational amplifier).

$$v_o = \left(1+\frac{R}{R}\right)v_1$$

Substitute the expression for v_1 from equation (1).

$$\begin{aligned} v_o &= -\left(1+\frac{R}{R}\right)(v_s+v_o) \\ &= -2(v_s+v_o) \\ &= -2v_s-2v_o \end{aligned}$$

Simplify this expression to get the following.

$$\begin{aligned} v_o+2v_o &= -2v_s \\ 3v_o &= -2v_s \end{aligned}$$

$$v_o = -\left(\frac{2}{3}\right)v_s$$

$$\frac{v_o}{v_s} = -\frac{2}{3}$$

Thus, the output of the circuit, v_o is $\boxed{-\frac{2}{3}}$.

Comments (1)

Anonymous

Please draw the current path on the diagram so that we know which resistor is Rf and which one is R1.